Ronin Dean Toy

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EDUCATION

San Diego State University, College of Engineering **May 2026**
B.S. in Mechanical Engineering San Diego, CA

SKILLS

Software: MATLAB, SolidWorks, AutoCAD, CREO, Arduino IDE, Simulink, Microsoft Office, Java, G-Code, TracerDAQ, InstaCal
Machining: GD&T (ASME Y14.5), HAAS CNC Mill, HAAS CNC Lathe, CNC Waterjet Cutter, MIG Welder, Flash Sintering Furnace, SHS Reactor, Bambu 3D Printers
Certifications: RCF Certification (CITI Programs)

EXPERIENCE

Lab Research Assistant **August 2025 - May 2026**
SDSU Advanced Materials Processing Lab (AMPL) San Diego, CA

- Conducted research on electrically driven spark plasma foaming of aluminum metal compacts, investigating the effect of applied current on heating rate and foam structure as an alternative to conventional furnace-based processing
- Ran systematic trials combining TiH_2 and Al powders across 250–1000A current outputs to optimize closed-cell aluminum foam structure; identified 750A as the optimal setting with a ~994°C/min heating rate and 67% total porosity
- Designed a Type-C thermocouple measurement setup and integrated it with existing Omega DAQ hardware and TracerDAQ software to capture real-time thermal and electrical data during high-current processing
- Performed root cause analysis on inconsistent foaming results, identifying heating rate variability as the primary driver and adjusting instrumentation and experimental approach accordingly
- Investigated the effect of green compact relative density on foaming behavior and heating response, testing samples across a 92–99% density range
- Performed Archimedes density analysis and cross-sectional porosity characterization on foamed samples to evaluate closed-cell structure quality and identify optimal processing parameters

Lead Engineer, Ember Generator Project **August 2025 - May 2026**
SDSU FLAMMA Lab San Diego, CA

- Led full development lifecycle of an ember generator for wildfire firebrand research, applying first principles analysis to redesign a prior iteration - replacing an auger-fed fuel system with a gravity-fed trapdoor mechanism, designing a straight pipe wind tunnel, and integrating a shaking mesh for continuous ember production
- Conducted rapid iterative prototyping using cardboard, sheet metal, and varied fuel types to validate design concepts before finalizing fabrication
- Assembled electromechanical control circuitry integrating Arduino, thermocouples, DC actuator, variable-heat hot rod, and variable-speed fan for automated combustion monitoring and control
- Coordinated with contractors, suppliers, sponsors, and team members through design, fabrication, and test phases
- Resolved a mid-project manufacturing non-conformance by reforming sheet metal components that came back outside weld tolerance; after an external vendor executed the weld incorrectly, redesigned the combustion chamber to integrate the shaking mechanism directly, eliminating the flawed weld interface and holding schedule
- Improved on the previous iteration by omitting overengineered components and applying DFM principles, resulting in an average 5x increase in ember generation, measured with an IR gun over a 30-second period
- Executed formal performance test & evaluation at a CAL Fire facility, running requirements verification against sponsor criteria and validating system performance across all seven design requirements
- Developed SolidWorks CAD drawings with GD&T-based tolerancing, a bill of materials for budget management, system-level diagrams, and technical reports and presentations delivered to the project sponsor

ACTIVITIES, AFFILIATIONS, AND AWARDS,

Student Organizations: San Diego Sales Engineering Club, Alpha Epsilon Pi Fraternity Construction Chair
Honors/Awards: Gold Level President's Volunteer Service Award (PVSA), International Baccalaureate Diploma